The Relationship of Sleep Quality, Stress Level, and Academic Performance among Chinese University Students

Abstract

Bad sleeping habits and stress are prevalent among Chinese university students because of their drive to excel academically. However, many studies state that poor sleep quality and stress could affect students' academics. A questionnaire regarding the three variables was conducted among CUHK-Shenzhen undergraduate students. The study reveals that 76% have poor quality sleep, and 80% show some signs of stress. However, academic performance with either sleep quality or stress shows no significant correlation. Therefore, Chinese university students with poor sleep quality and stress can still maintain a solid academic performance.

I. Introduction

Sleep, connected to health and wellbeing, is a basic requirement for humans (Cates et al., 2015). Feeling energized after sleeping is characterized as the sleep quality (Karatay et al., 2016). However, currently, concerns about sleep deprivation have been rising, and poor sleep quality has become an indication of medical disorders (Akhlaghi A.A. et al., 2009). Sleep deprivation has become a common health concern among university students. It has been observed that sleep quality of university students has been deteriorating (Vail-Smith et al., 2009). Furthermore, stress as an inevitable aspect of life is regularly encountered by university student. Students endure several psychological issues in their new surroundings, compromising their well-being (Kitzrow, 2003); This happens because the university system has a considerable difference in academic standards. In conclusion, stress is quite prevalent in university students' life since they need to assure their academic survival and prepare for a future employment.

Very few research in China have examined the link between sleep quality, stress levels, and academic performance of university students. A meta-analysis of 57 publications indicated that short sleep duration and irresponsible sleeping practices are frequent among Chinese university students (Li et al., 2017). Multiple studies have also demonstrated significant distress among Chinese university students (Wang et al., 2021; Wang et al., 2020; Cao et al., 2020). However, Chinese students have exhibited superior academic

achievement than other overseas students (Li et al., 2010). This study intends to investigate the association between sleep quality, stress level, and academic performance of Chinese university students.

II. Literature review

Sleep is key in human physiology because sleep promotes everyday functioning. However, over 70 million individuals in America alone suffer from sleep deprivation. Sleep deprivation's adverse impacts on human include deteriorating cognitive capabilities (Colten et al., 2006). Every student's cognitive functioning is vital for their academics. However, sleep loss may significantly impair students' capacity to focus (Pilcher et al., 2011). A research has indicated that 55.8% of university students in Ethiopia suffer from poor sleep quality outcomes (Lemma et al., 2012).

Sleep indicates significant relevance for increasing the working memory ability (Fenn et al., 2011). A meta-analytical study of 70 articles, containing sleep deprivation test results, concluded that sleep deprivation harms humans' cognitive domains such as simple attention, complex attention, working memory, processing speed, short-term memory, and reasoning (Lim et al., 2010). A research regarding the correlation between sleep quality and students' academic says that poor sleep quality during exam periods may negatively affect students' test outcomes (Ahrberg et al., 2012). Moreover, stress, another element most students confront, impairs sleep quality. A study has indicated that a high amount of stress that students face everyday may significantly effect their academic outcomes (Sohail, 2013).

Although there is abundant evidence on how sleep quality and stress impact a student's academics, there is still not enough study that explores the association between the three variables. Therefore, this study will address that gap by investigating the association between sleep quality, stress, and academic performance. The research question is "What is the association between sleep quality, stress level, and academic performance among Chinese university students?".

III. Method

A. Participants

The participants of this study were 50 undergraduate university students in CUHK-Shenzhen.

B. Materials

An online questionnaire was used to collect data. The questionnaire was divided into three sections. The first section records the sleep quality using the Pittsburgh Sleep Quality Index (PSQI) test (Buysse et al., 1989). The PSQI score ranges from 0 to 21, and a score of 1-7 indicates good sleep quality and > 8 indicates otherwise. The second section records the stress level using the Kessler Psychological Distress Scale (K10) test (Kessler et al., 2002). The K10 score ranges from 10 to 50, and a score of <20 indicates well-being, 20-24 indicates mild stress, 25-29 indicates moderate stress, and >30 indicates severe stress. Finally, the third section records the student’s GPA.

C. Procedure

The data were then analyzed using Python (version 3.9). The Pearson Correlation Coefficient was calculated to understand the correlation between each variable. The coefficient (r) ranges from -1 to 1, with -1 indicating a strong negative correlation, 0 indicating no correlation, and 1 indicating a strong positive correlation (Buchanan et al., 2012, p.359) . The correlation was then plotted in linear regression modes.

IV. Results

A. Findings

The characteristics of the studied participants are shown in Table 1. Most students were considered very good for their GPA (78.0%; GPA: 3.1-3.5) and some students were considered excellent (12.0%; GPA: >3.5); and about one-tenth were below average (10.0%; GPA: 0-3.0). The prevalence of poor sleep quality of the participants was 76% (n = 38, PSQI score: >8). Stress was also prevalent: 82% (n = 41) of students showed some sign of stress. 76% (n = 38) of students had moderate or severe stress levels. Mild stress level and well-being were 18% (n = 3) and 6% (n = 3) respectively.

Table 1: Characteristics of Participants

Characteristics	n	%
GPA		
≤ 3.0	6	12.0%
3.1– 3.5	39	78.0%
> 3.5	5	10.0%
Poor Sleep Quality		
Yes	38	76.0%
No	12	24.0%
Stress level		
Well	9	18.0%
Mild	3	6.0%
Moderate	29	58.0%
Severe	9	18.0%

Note: n refers to the number of students

B. Analysis

Figure 1 shows a heatmap correlation between PSQI, K10, and GPA scores. From Figure.1, it is shown that sleep quality (PSQI) and level of distress (K10) have a strong positive correlation (r = 0.61). However, the correlation between GPA with sleep quality and stress level shows no significant correlation, with the coefficient being 0.1 and -0.2, respectively. The correlation between variables is plotted in a linear regression model (Figure 2).

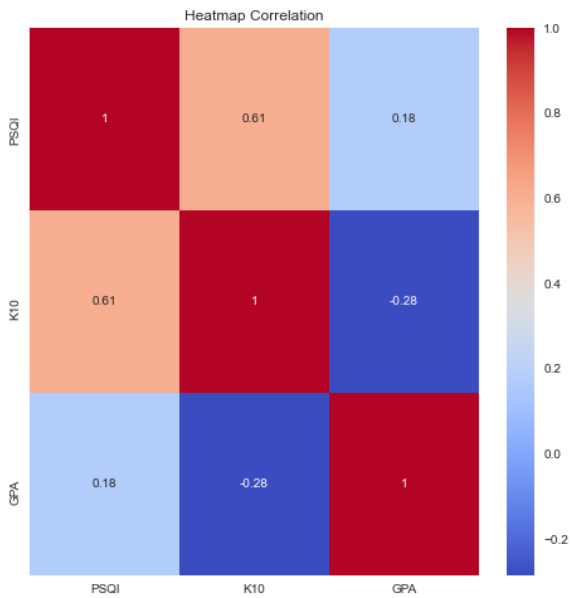

Figure 1: Heatmap Correlation

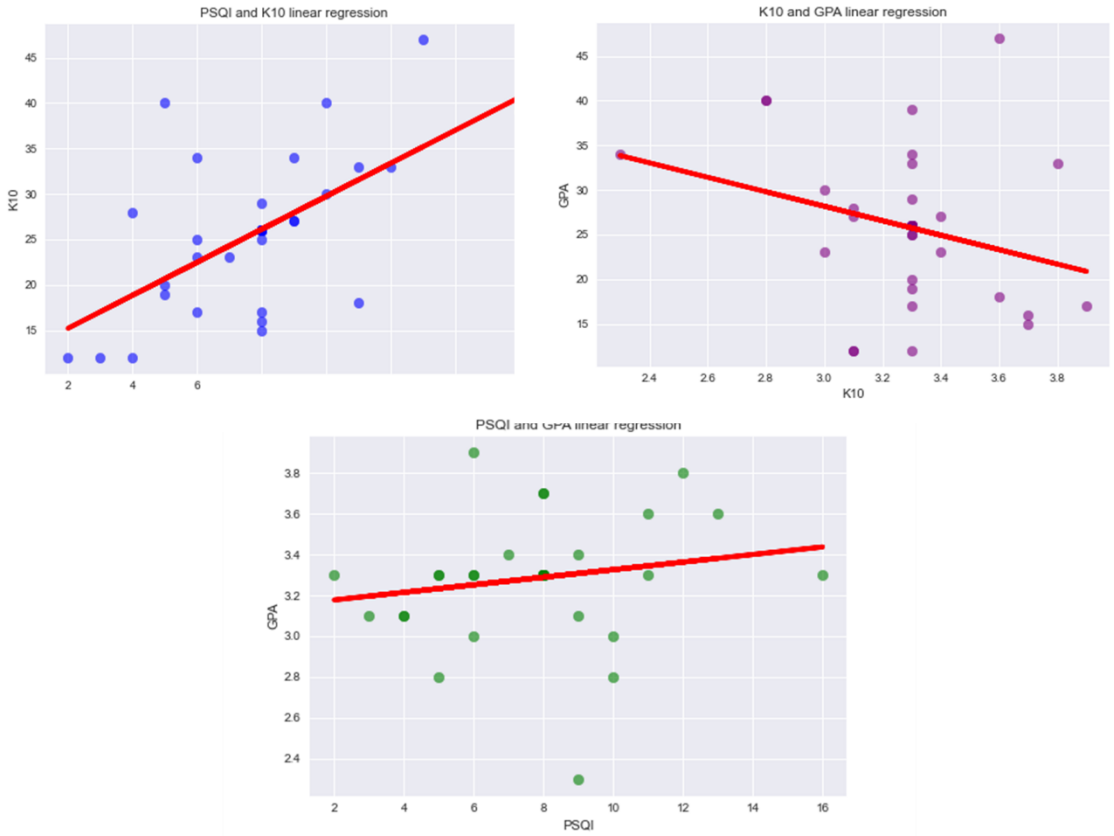

Figure 2: Linear regression models

V. Discussion

In the research, 76% of participants reported indicators of poor sleep quality. This conclusion is similar to a meta-analytical research of sleep habits of Chinese university students (52.3%) (Li et al., 2017). Globally, the quality of sleep of university students was found to be better: 61.7% in Jordan (Toubasi et al., 2021), 55.8% in Ethiopia (Lemma et al., 2012), and 33.8% in India (Kaur, 2018). The prevalence of stress among participants in this study is 76%, also greater than in several international studies. For instance, a frequency of 71% was discovered in a university in the United States (Son et al., 2020) and 53% in Australia (Lee et al., 2016). Students' stress levels and poor sleep quality elevate during pre-examination period (Ahrberg et al., 2012); This might explain the high prevalence of poor sleep quality and stress given most final examinations are held at thebeginning of December.

Poor sleep quality revealed a strong link with the degree of stress in this study (r = 0.61). Numerous research have widely shown this association (Zunhammer et al., 2014; Ahrberg et al., 2012). A research done by Morin et al. (2003) states that an increase in stressors and the lack of control over stressful events would boost the sensitivity to different sleep disorders, including insomnia. He also argues that there is a direct association between insomnia and stress. However, this study found an insignificant link between the

relationship of academic performance with either sleep quality or stress level; this may reflect that students were already well-adapted to function well academically under poor sleep quality and high distress.

This study is based on one university, thus results from other institutions in China can change because every school has a distinct structure. The subjects of this research comprise students from all various disciplines. Therefore, students may differ in their workload, course complexity, and cognitive approach. Future research are recommended to study each student's behavioral and psychological features. Retrieving data concerning caffeine consumption and midday naps might be a fantastic supplementary variable. Meanwhile, sleep quality and stress levels in China are encourage to be more addresed.

VI. Conclusion

The present study has explored the relationship between sleep quality, stress level, and academic performances among Chinese university students. The findings reinforce and improve those reported in previous studies such as Ahrberg et al. (2012) and Sohail (2013). The study enriches a more profound understanding of sleeping habits and stress levels and how it associates with academics by analyzing their correlation. The limitation of the study includes not collecting enough variables and not targeting a specific major. Future researchers are encouraged to collect more evidence from students that might affect their academic life.

VII. References

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VIII. Appendix

Table 2: Raw Data

PSQI	K10	GPA
8	26	3.3
16	39	3.3
8	26	3.3
6	23	3
8	26	3.3
8	26	3.3
8	25	3.3
8	26	3.3
8	26	3.3
3	12	3.1
9	27	3.4
8	26	3.3
6	34	3.3
5	20	3.3
10	30	3
8	26	3.3
8	26	3.3
8	16	3.7
8	15	3.7
10	40	2.8
8	26	3.3
8	26	3.3
7	23	3.4
8	26	3.3
11	33	3.3
9	34	2.3
8	26	3.3
6	17	3.9
8	26	3.3
8	26	3.3
8	26	3.3
8	29	3.3
8	26	3.3
9	27	3.1
5	40	2.8
8	26	3.3
13	47	3.6
4	28	3.1
8	26	3.3
8	26	3.3
11	18	3.6
8	26	3.3
8	26	3.3
8	26	3.3
2	12	3.3
5	19	3.3
6	25	3.3

8	17	3.3
4	12	3.1
12	33	3.8

Python code

December 5, 2021

1 Packages

[30]:
```
import pandas as pd
from matplotlib import pyplot as plt
import numpy as np
import seaborn
from sklearn import linear_model
from matplotlib import style
%matplotlib inline
```

2 Raw data

[31]:
```
df = pd.read_csv('results.csv')
df.drop(columns=['Unnamed: 3', 'Unnamed: 4', 'Unnamed: 5', 'Unnamed: 6',␣
 ↪'Unnamed: 7'], inplace=True)
df
```

[31]:

	PSQI	K10	GPA
0	8	26	3.3
1	16	39	3.3
2	8	26	3.3
3	6	23	3.0
4	8	26	3.3
5	8	26	3.3
6	8	25	3.3
7	8	26	3.3
8	8	26	3.3
9	3	12	3.1
10	9	27	3.4
11	8	26	3.3
12	6	34	3.3
13	5	20	3.3
14	10	30	3.0
15	8	26	3.3
16	8	26	3.3
17	8	16	3.7
18	8	15	3.7

```
19    10   40  2.8
20     8   26  3.3
21     8   26  3.3
22     7   23  3.4
23     8   26  3.3
24    11   33  3.3
25     9   34  2.3
26     8   26  3.3
27     6   17  3.9
28     8   26  3.3
29     8   26  3.3
30     8   26  3.3
31     8   29  3.3
32     8   26  3.3
33     9   27  3.1
34     5   40  2.8
35     8   26  3.3
36    13   47  3.6
37     4   28  3.1
38     8   26  3.3
39     8   26  3.3
40    11   18  3.6
41     8   26  3.3
42     8   26  3.3
43     8   26  3.3
44     2   12  3.3
45     5   19  3.3
46     6   25  3.3
47     8   17  3.3
48     4   12  3.1
49    12   33  3.8
```

[32]:
```
df[(df['GPA'] <= 3.0)].count()
```

[32]:
```
PSQI    5
K10     5
GPA     5
dtype: int64
```

3 Statistics

[33]:
```
bins = [-1, 7, 14, np.inf]
labels=['1-7','8-14','>14']
df_PSQI = df.groupby(pd.cut(df['PSQI'], bins=bins, labels=labels)).size().
 ↪reset_index(name='count')
df_PSQI['%'] = (df_PSQI['count'] / 50) * 100
df_PSQI
```

[33]:

	PSQI	count	%
0	1-7	12	24.0
1	8-14	37	74.0
2	>14	1	2.0

[34]:
```
df[df['PSQI'] > 8].mean()
```

[34]:
```
PSQI    11.00
K10     32.80
GPA      3.22
dtype: float64
```

[35]:
```
bins = [-1, 19, 24, 29, np.inf]
labels=['<20','20-24','25-29', '>30']
df_K10 = df.groupby(pd.cut(df['K10'], bins=bins, labels=labels)).size().
 ↪reset_index(name='count')
df_K10['%'] = (df_K10['count'] / 50) * 100
df_K10
```

[35]:

	K10	count	%
0	<20	9	18.0
1	20-24	3	6.0
2	25-29	29	58.0
3	>30	9	18.0

[36]:
```
bins = [-1, 3.0, 3.5, np.inf]
labels=['0-3.0','3.1-3.5','>3.5']
df_GPA = df.groupby(pd.cut(df['GPA'], bins=bins, labels=labels)).size().
 ↪reset_index(name='count')
df_GPA['%'] = (df_GPA['count'] / 50) * 100
df_GPA
```

[36]:

	GPA	count	%
0	0-3.0	5	10.0
1	3.1-3.5	39	78.0
2	>3.5	6	12.0

[37]:
```
df.describe()
```

[37]:

	PSQI	K10	GPA
count	50.000000	50.000000	50.000000
mean	7.860000	25.860000	3.286000
std	2.347469	6.995655	0.245789
min	2.000000	12.000000	2.300000
25%	8.000000	25.000000	3.300000
50%	8.000000	26.000000	3.300000
75%	8.000000	26.750000	3.300000
max	16.000000	47.000000	3.900000

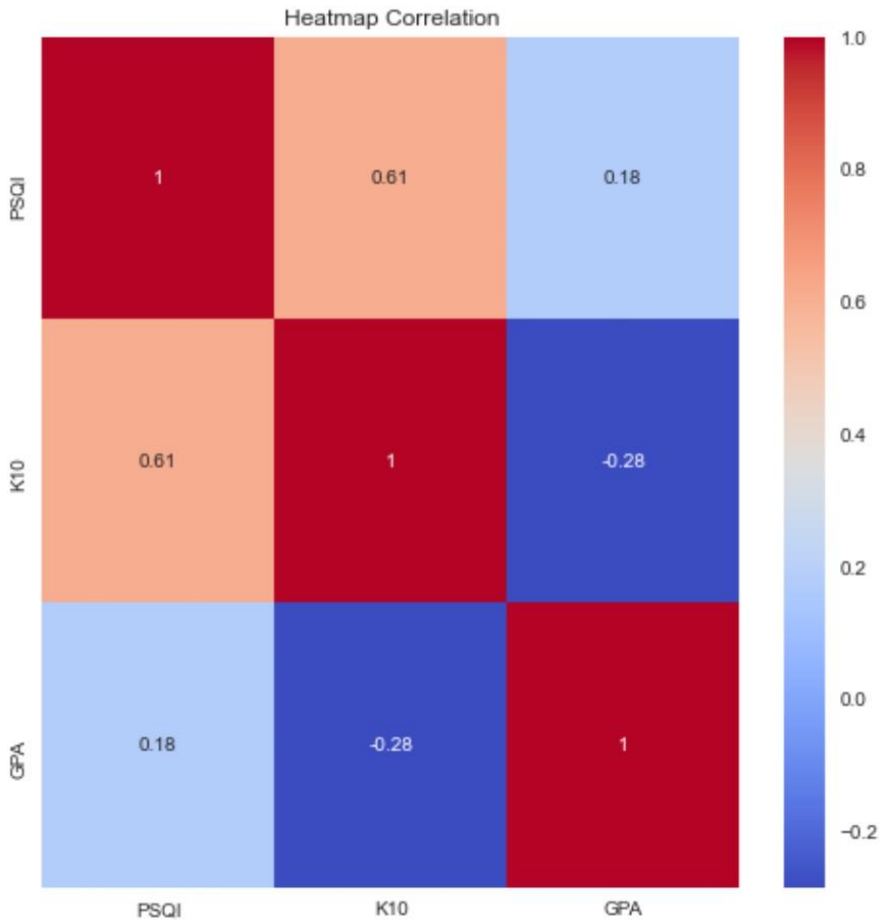
4 Pearson Correlation

[38]:
```
df.corr()
```

[38]:

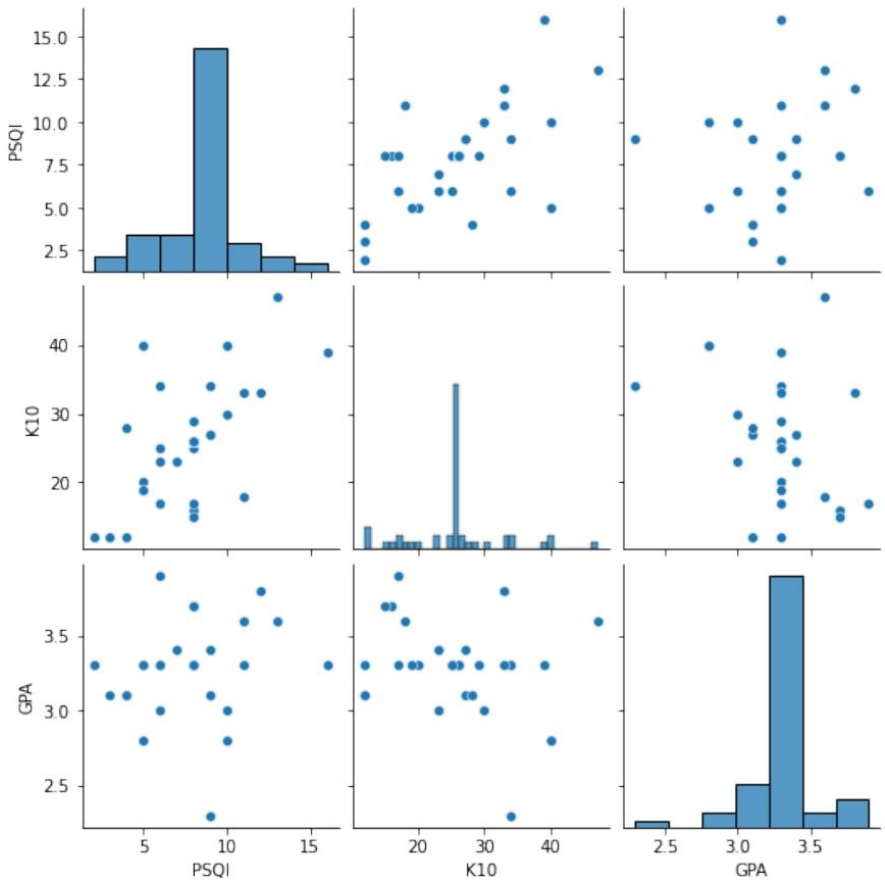
	PSQI	K10	GPA
PSQI	1.000000	0.610204	0.176923
K10	0.610204	1.000000	-0.284832
GPA	0.176923	-0.284832	1.000000

[45]:
```
plt.figure(figsize=(8,8))
seaborn.heatmap(df.corr(), annot=True, cmap="coolwarm").set_title('Heatmap␣
↪Correlation')
plt.savefig('heatmap_corr.png')
```



[40]:
```
seaborn.pairplot(df)
```

[40]: <seaborn.axisgrid.PairGrid at 0x226c2937fa0>



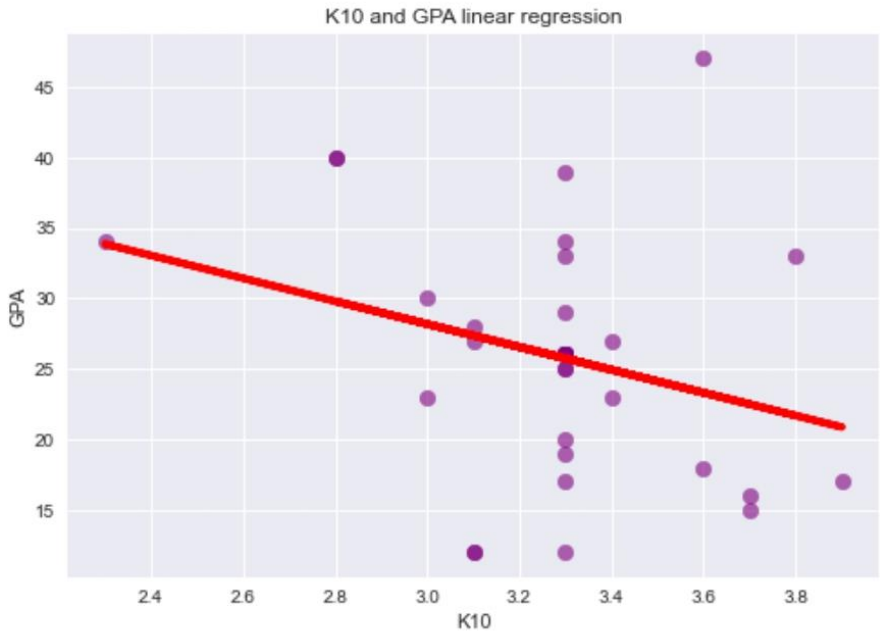
5 Linear Regression

[47]:
```
PSQI_list = df['PSQI'].tolist()
K10_list = df['K10'].tolist()

xs = np.array(PSQI_list, dtype=np.float64)
ys = np.array(K10_list, dtype=np.float64)

PSQI_K10 = linear_model.LinearRegression()
PSQI_K10.fit(xs.reshape(-1,1), ys)
regression_line = PSQI_K10.predict(xs.reshape(-1,1))
```

```
style.use('seaborn')
plt.scatter(xs, ys, label='Data Points', alpha=0.6, color='blue', s=75)
plt.plot(xs, regression_line, label='Best Fit Line', color='red', linewidth=4)
plt.title('PSQI and K10 linear regression')
plt.xlabel('PSQI')
plt.ylabel('K10')
plt.savefig('foo.png')
plt.show()
```

[49]:
```
PSQI_list = df['PSQI'].tolist()
GPA_list = df['GPA'].tolist()

xs = np.array(PSQI_list, dtype=np.float64)
ys = np.array(GPA_list, dtype=np.float64)

PSQI_GPA = linear_model.LinearRegression()
PSQI_GPA.fit(xs.reshape(-1,1), ys)
regression_line = PSQI_GPA.predict(xs.reshape(-1,1))

style.use('seaborn')
plt.scatter(xs, ys, label='Data Points', alpha=0.6, color='green', s=75)
```

```
plt.plot(xs, regression_line, label='Best Fit Line', color='red', linewidth=4)
plt.title('PSQI and GPA linear regression')
plt.xlabel('PSQI')
plt.ylabel('GPA')
plt.savefig('psqi_gpa.png')
plt.show()
```

[50]:
```
GPA_list = df['GPA'].tolist()
K10_list = df['K10'].tolist()

xs = np.array(GPA_list, dtype=np.float64)
ys = np.array(K10_list, dtype=np.float64)


K10_GPA = linear_model.LinearRegression()
K10_GPA.fit(xs.reshape(-1,1), ys)
regression_line = K10_GPA.predict(xs.reshape(-1,1))

style.use('seaborn')
plt.scatter(xs, ys, label='Data Points', alpha=0.6, color='purple', s=75)
plt.plot(xs, regression_line, label='Best Fit Line', color='red', linewidth=4)
```

```
plt.title('K10 and GPA linear regression')
plt.xlabel('K10')
plt.ylabel('GPA')
plt.savefig('k10_gpa.png')
plt.show()
```